

Lesson 07 Practice: Computer Memory Hierarchy

JKSSB Computer Awareness

October 10, 2026

Provenance: All 25 questions are original exam-pattern questions (*[Original]* plus the *[JKSSB FAA-pattern]*, *[JKPSI-pattern]*, *[SSC CGL-pattern]*, and *[JKSSB-pattern]* extension items, including MSQs); none reproduces an official paper item.

Q1. *[Original]* Expand RAM.

- A. Read Access Memory
- B. Random Access Memory
- C. Rapid Access Memory
- D. Random Allocation Memory

Q2. *[Original]* Expand ROM.

- A. Random-Only Memory
- B. Read-Once Memory
- C. Read-Only Memory
- D. Runtime Object Memory

Q3. *[Original]* Which option correctly lists the memory hierarchy in order of decreasing speed (fastest first)?

- A. Registers, cache, main memory (RAM), secondary storage
- B. Cache, registers, main memory (RAM), secondary storage
- C. Main memory (RAM), registers, cache, secondary storage
- D. Secondary storage, main memory (RAM), cache, registers

Q4. *[Original]* Which of the following is the fastest storage in a computer system?

- A. Main memory (RAM)
- B. L1 cache
- C. Solid-state drive (SSD)
- D. Registers

Q5. *[Original]* Cache memory is _____ than main memory (RAM).

- A. slower
- B. faster

- C. larger
- D. cheaper per bit

Q6. [Original] Which statement correctly distinguishes cache memory from main memory (RAM)?

- A. Cache is larger and faster than RAM.
- B. Cache and RAM have the same speed and size.
- C. Cache is smaller and faster than RAM; RAM is larger and slower.
- D. Cache is a permanent backup of the contents of RAM.

Q7. [Original] Which level of cache is the smallest and fastest?

- A. L1
- B. L2
- C. L3
- D. L4

Q8. [Original] Among the cache levels L1, L2, and L3, which is the largest and slowest (though still faster than main memory)?

- A. L1
- B. L2
- C. Registers
- D. L3

Q9. [Original] Which of the following is **NOT** true of cache memory?

- A. It stores recently and frequently used data.
- B. It helps exploit locality of reference.
- C. It acts as a permanent backup for the contents of RAM.
- D. It is faster than main memory.

Q10. [Original] RAM is a _____ memory, whereas ROM is _____.

- A. non-volatile; volatile
- B. volatile; non-volatile
- C. read-only; writable
- D. secondary; primary

Q11. [Original] Which memory loses its contents when the power is switched off?

- A. RAM
- B. ROM
- C. Hard disk drive
- D. Solid-state drive

Q12. [Original] Which of the following is **primary memory**?

- A. Solid-state drive
- B. Hard disk drive
- C. Magnetic tape
- D. RAM

Q13. *[Original]* Which of the following is a **secondary storage** device?

- A. Register
- B. Cache
- C. Solid-state drive (SSD)
- D. RAM

Q14. *[Original]* Why does a memory hierarchy exist?

- A. Faster memory is smaller and more expensive per bit, so the hierarchy balances speed, cost, and capacity.
- B. All memory types have the same speed and cost.
- C. Secondary storage is the fastest type of memory.
- D. Registers can store the entire contents of main memory.

Q15. *[Original]* Evaluate the statements:

- I. Registers are the fastest storage in a computer.
- II. Cache memory is faster than main memory.
- III. RAM is non-volatile and retains its data when the power is off.

- A. I only
- B. I and II only
- C. II and III only
- D. I, II, and III

Q16. *[Original]* The tendency of a processor to reuse recently accessed data and nearby data is called:

- A. locality of reference
- B. overflow
- C. fragmentation
- D. deadlock

Q17. *[Original]* Temporal locality refers to:

- A. accessing data stored at nearby addresses.
- B. storing data permanently on secondary storage.
- C. reusing the same data or instruction again soon after it was accessed.
- D. losing data when the power is switched off.

- Q18.** *[Original]* Which memory is typically the **smallest** in capacity?
- A. Solid-state drive (SSD)
 - B. Main memory (RAM)
 - C. L1 cache
 - D. Registers
- Q19.** *[Original]* Which component retains its contents without power, so it can store files between sessions?
- A. Registers
 - B. Secondary storage such as an SSD or HDD
 - C. Cache memory
 - D. RAM
- Q20.** *[Original]* Which pairing is correctly matched?
- A. Registers — slowest storage in the computer
 - B. Cache — non-volatile permanent backup of RAM
 - C. RAM — non-volatile read-only memory
 - D. SSD — secondary storage that retains data without power
- Q21.** *[JKSSB FAA-pattern]* Which memory in the hierarchy is the fastest?
- A. Registers
 - B. Hard disk
 - C. Magnetic tape
 - D. DVD
- Q22.** *[JKPSI-pattern]* Which option lists memories in decreasing order of speed (fastest first)?
- A. Registers, cache, RAM, hard disk
 - B. Hard disk, RAM, cache, registers
 - C. RAM, hard disk, registers, cache
 - D. Cache, hard disk, registers, RAM
- Q23.** *[SSC CGL-pattern]* Which memory sits between the CPU and main memory to speed up access?
- A. Cache memory
 - B. Magnetic tape
 - C. Floppy disk
 - D. Printer buffer card
- Q24.** *[JKSSB-pattern]* (**MSQ: select ALL correct options.**) Which of the following are primary (main) memory?

- A. RAM
 - B. ROM
 - C. Cache memory
 - D. Hard disk drive
- Q25.** *[SSC CGL-pattern]* (MSQ: select ALL correct options.) Which statements about the memory hierarchy are correct?
- A. Registers are the fastest storage in the hierarchy.
 - B. Cache speeds up access to frequently used data.
 - C. Secondary storage is non-volatile.
 - D. ROM is volatile working memory.
- Q26.** *[PYQ-adapted: JKSSB Junior Assistant 2026 Q63, final key; SRC-03/SRC-06]* Which memory is typically faster than main memory and holds copies of recently used data?
- A. Cache
 - B. HDD
 - C. Optical disc
 - D. Magnetic tape
- Q27.** *[Original]* In the usual speed hierarchy, which is fastest?
- A. Registers
 - B. RAM
 - C. SSD
 - D. Magnetic tape
- Q28.** *[Original]* As capacity increases and cost per bit generally falls, memory in a computer is usually:
- A. Faster at every level
 - B. Slower at each lower level of the hierarchy
 - C. Volatile at every level
 - D. Removed from the CPU
- Q29.** *[Original]* Which pair is correctly ordered from faster to slower?
- A. RAM, registers, cache
 - B. Registers, cache, RAM
 - C. SSD, RAM, registers
 - D. Tape, cache, registers
- Q30.** *[Original, MSQ drill]* (MSQ: select ALL correct options.) Which statements describe the memory hierarchy?
- A. Registers are very small and fast.
 - B. Cache can reduce average access time.
 - C. Secondary storage is generally non-volatile.
 - D. Magnetic tape is faster than CPU registers.